

Erdős Problem #151

EP-151

L1 - TRACTABLE

OPEN

Graph Theory

ORIGINAL PROBLEM STATEMENT

For a graph G let $\tau(G)$ denote the minimal number of vertices that include at least one from each maximal clique of G on at least two vertices (sometimes called the clique transversal number).

Let $H(n)$ be maximal such that every triangle-free graph on n vertices contains an independent set on $H(n)$ vertices.

If G is a graph on n vertices then is $\tau(G) \leq n - H(n)$?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>